SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	1 - Introduction & Basic Concepts

1. Introduction to Surveying

Surveying is the art and science of determining the relative positions of points on, above or beneath the surface of the earth by measuring horizontal distances, vertical distances (elevations) and angles, and then plotting them to scale as a plan or map.

- **Plan:** a horizontal projection of the area drawn to a large scale (small areas, e.g. 1 cm = 5 m).
- **Map:** a drawing of a large area drawn to a small scale (e.g. 1 cm = 1 km).
- Surveying is the first step of every civil engineering project - roads, railways, canals, dams, buildings, bridges, pipelines.
- Measurements are made in either the **horizontal plane** (distances, angles, bearings) or the **vertical plane** (levels, heights).

Primary divisions

- **Plane surveying:** earth surface taken as a plane; curvature ignored. Used for areas up to about 260 sq.km.
- **Geodetic surveying:** curvature of the earth is considered. Used for large areas and high accuracy (more than 260 sq.km).

2. Objectives of Surveying

- To prepare a **topographic map** showing natural and man-made features - hills, rivers, roads, towns.
- To prepare a **cadastral map** showing field and property boundaries, used for land records and taxation.
- To prepare **engineering maps** for roads, railways, canals, dams etc.; to fix the alignment of the project.

- To calculate **areas and volumes** of land and earthwork (cutting and filling).
- To **set out** (mark on ground) the proposed structure from its plan – this is called setting out or layout.
- To collect data for estimating cost, and to resurvey and check existing works.

3. Classification of Surveying

(a) Based on the nature of the field

- **Land surveying:** topographical, cadastral, city and engineering surveys.
- **Marine / hydrographic surveying:** water bodies – tides, depth (soundings), shore lines.
- **Astronomical surveying:** positions of heavenly bodies to fix latitude, longitude and azimuth.
- **Aerial surveying:** photographs taken from aircraft or drones (see Unit V).

(b) Based on the instrument used

Survey	Main instrument	Used for
Chain survey	Chain / tape, cross staff	Small, open, flat areas
Compass survey	Prismatic compass	Bearings of lines, traverse
Plane table survey	Plane table, alidade	Plotting in the field itself
Levelling	Dumpy / tilting level, staff	Elevations, profiles
Theodolite survey	Theodolite	Precise angles, traverse
Tacheometry	Tacheometer	Rapid horizontal / vertical dist.
Total station / GPS	EDM, GNSS receiver	Modern high-precision work

(c) Based on the method

- **Triangulation:** control points form a network of triangles; one baseline and angles are measured.
- **Traversing:** a series of connected lines whose lengths and directions are measured.

(d) Based on the purpose

- Engineering survey (reconnaissance, preliminary, location), military survey, mine survey, archaeological survey, geological survey.

4. Principles of Surveying

KEY PRINCIPLES:

Two fundamental principles: (1) Work from the whole to the part. (2) Locate a new station by at least two independent measurements from fixed reference points.

Principle 1: Working from whole to part

First establish a major **control** (a framework of well-spaced, accurately measured main stations). Then fill in the minor details inside this frame using less accurate methods.

- Reason: errors do not accumulate; any error in small details stays confined to that portion.
- The reverse (part to whole) lets errors build up and the final result gets distorted.

Principle 2: Fixing a new point

A new station can be fixed from two reference points A and B by any one of these:

- Measuring two distances (linear measurements) from A and B.
- Measuring one distance and one angle (or bearing).
- Measuring two angles (or bearings) - intersection.
- Measuring distance to the new point from A and a perpendicular offset from the line AB.

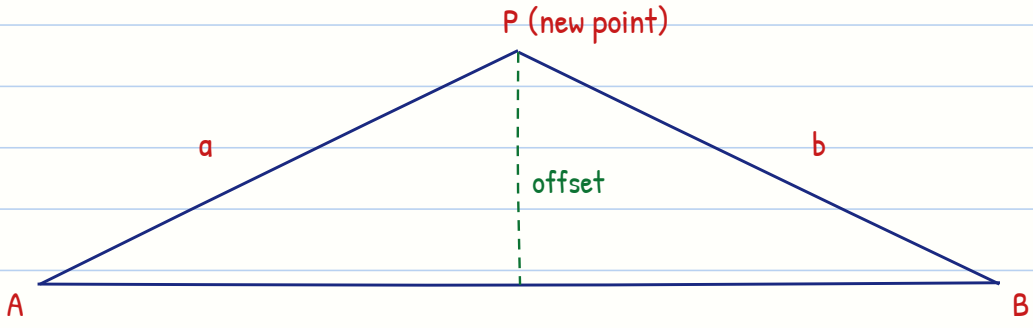


Fig 1: fixing a point P from baseline AB

Other general principles: (3) Check lines and tie lines must always be provided to detect errors. (4) Accuracy of the work should suit the purpose and scale (precision should not exceed what the drawing can show).

5. Surveying Accessories

Accessory	Purpose
Chain / Tape	Direct measurement of linear distance
Arrows (marking pins)	10 per set; mark every chain length on ground

Ranging rods	2-3 m poles, painted red-white 0.2 m bands; to mark stations and align the line
Ranging poles	Like rods but 3-6 m long, for long lines
Offset rod	3 m rod with hook, used to measure offsets
Cross staff	Sets out right angles (offsets) - open, French, adjustable
Optical square	Sets right angles using two mirrors at 45 deg
Plumb bob	Transfers points vertically; plumbing a staff
Pegs	Wooden, mark station points; 2.5 cm sq, 15 cm long
Levelling staff	Reading of levels with a level

6. Introduction to Compass Surveying

A compass survey fixes the ***direction (bearing)*** of survey lines using a magnetic needle. Lengths are measured with chain or tape. It is suited to rough, rapid work on large areas, rugged or wooded country.

- Instruments: ***prismatic compass*** (more common) and surveyor's compass.
- Method: open ***traverse*** (not closed) or ***closed traverse*** . Fore bearing and back bearing of each line are taken.
- Accuracy: angular accuracy about 15 minutes; not suitable near steel structures, power lines.

7. Introduction to Levelling

Levelling is the branch of surveying that determines the relative heights (elevations) of points above or below a reference surface called the ***datum*** (usually Mean Sea Level).

- Instruments: dumpy level, tilting level, auto level, levelling staff.
- Uses: preparing longitudinal and cross sections, contouring, fixing grades and formation levels for roads, canals, drains.
- Reduced Level (R.L.) = elevation of a point above datum. Detailed study in Unit II.

8. Introduction to Plane Table Surveying

In ***plane table surveying*** fieldwork and plotting are done simultaneously. A drawing board (plane table) is mounted on a tripod and a sheet of paper fixed on it. Points are sighted through an ***alidade*** and rays are drawn directly.

- Accessories: plane table with tripod, alidade (plain or telescopic), spirit level, plumbing fork with plumb bob, trough compass, drawing sheet, pins.
- Methods: ***radiation, intersection, traversing, resection*** .
- Advantages: no field book needed, mistakes and omissions noticed on the spot, suited for small scale maps.
- Disadvantages: not suitable in wet or windy weather; needs more carrying equipment; less accurate.

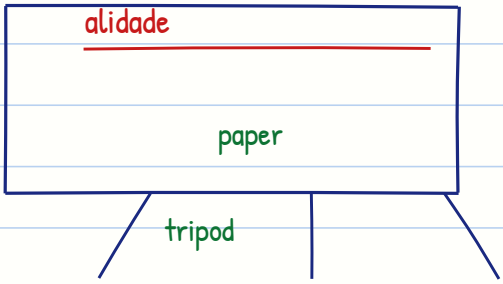


Fig 2: plane table on tripod

9. Linear Distances – Approximate Methods

Used for reconnaissance and checking of precise methods:

- ***Pacing:*** number of paces x average pace length (about 0.75 m). Accuracy about 1 in 100. Pace length is found by walking a known distance.
- ***Passometer:*** a watch-like device that counts steps; distance = steps x step length.
- ***Pedometer:*** records distance directly for a set stride length.
- ***Odometer / speedometer:*** attached to a wheel or vehicle; distance = wheel turns x circumference. Used on roads.
- ***Measuring wheel (perambulator):*** wheel pushed along the ground; dial shows distance.
- ***Chain-less estimation:*** by time taken (speed x time), or by sound/optical range finders.
- ***Tacheometric / EDM*** methods are accurate and are covered separately.

10. Direct Methods of Measuring Distance

Distance is directly measured on the ground by chains or tapes.

Types of chains

Chain	Length	No. of links	Remarks
Metric chain	20 m or 30 m	100 or 150 (20 m = 100)	Link = 0.2 m; tallies at every 1 m
Gunter's chain	66 ft	100	Link = 7.92 in; 10 chains = 1 furlong
Engineer's chain	100 ft	100	Link = 1 ft
Revenue chain	33 ft	16	Used for cadastral survey (land records)

- Metric chain: tallies – ***brass tags*** – used for easy reading: 1 tooth = 2 m, 2 teeth = 4 m, 3 teeth = 6 m, 4 teeth = 8 m; centre tag (10 m) is round.
- Chain is made of galvanised mild steel wire, 4 mm dia; links bent at both ends and joined with three small rings; brass handles at ends.
- ***Length of chain*** is measured from outside of one handle to outside of the other.

Chain surveying: principle and basics

- Principle: ***triangulation*** – the area is divided into well-conditioned triangles; sides are measured, details by offsets.
- ***Base line*** = longest main line; ***check lines / proof lines*** verify accuracy; ***tie lines*** fix intermediate details.
- Well-conditioned triangle: no angle less than 30 deg or more than 120 deg (ideally equilateral).

Tapes

Tape	Material / features
Cloth or linen tape	Linen with brass wires; 10, 20, 30 m; stretches and shrinks – only for offsets
Metallic tape	Linen with fine brass/copper wires; more durable
Steel tape	Steel band 6–16 mm wide, 30 or 50 m; accurate; rust-prone
Invar tape	36% nickel + 64% steel alloy; very low thermal expansion; for base line (highest accuracy)

11. Ranging

Ranging is the process of establishing intermediate points on a straight line between two end stations so that the chain can be laid in a straight line.

Direct ranging

Intermediate points are visible from end stations. Ranging rods are fixed at A and B. The assistant at C (near A) is directed by the surveyor standing 2 m behind A and sighting A–B until all rods are in line.

Indirect (reciprocal) ranging

Used when end stations are not intervisible due to a rise or obstacle. Two chainmen M and N stand with rods at intermediate points, each in sight of the other's rod and one end. They direct each other alternately until both rods are in line with A and B.

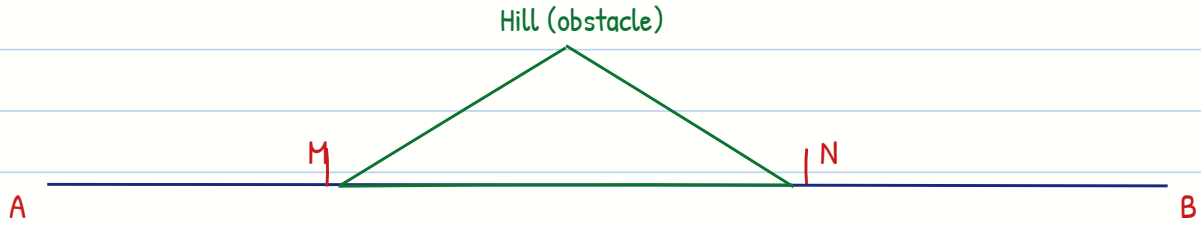


Fig 3: reciprocal ranging over a rise

12. Chaining a Line and Offsets

- The **leader** carries arrows and rods; the **follower** holds the rear handle and directs the leader on line.
- At the end of each chain the leader plants an arrow; the follower collects it. Arrow count in follower's hand = chains measured.
- **Offsets** are lateral measurements from the chain line to a feature. Perpendicular offset: set with cross staff or optical square. Oblique offset: measured at an angle (tie line).
- Limit: perpendicular offset up to 15 m (ordinary) and oblique offsets 20 m with chain; use of tape allows longer.

13. Errors in Chaining and Tape Corrections

Corrections are applied when the tape is not of standard length or conditions differ from standard. Errors may be cumulative (+/-) or compensating.

(a) Correction for wrong length (incorrect chain)

True length = L' x (L + e) / L

Correction Ce = L x e / L' (for measured length l: Ce = l x e / L)

e = change in chain length

- If the chain is **too long** , measured distance is too short -> **add** correction.
- If the chain is **too short** , measured distance is too long -> **subtract** correction.
- Area correction: True area = measured area x (L'/L)^2.

(b) Correction for temperature

Ct = alpha x (Tm - To) x L

alpha = coefficient of expansion (steel: 11.2 x 10^-6 per deg C)

- If Tm > To (standard temp.), tape lengthens: correction is **added** ; otherwise subtracted.

(c) Correction for pull (tension)

Cp = (P - Po) x L / (A x E)

P = applied pull, Po = standard pull, A = area of cross-section, E = Young's modulus

(d) Correction for sag (tape suspended between supports)

Cs = w^2 x l^3 / (24 x P^2) per span, or (W^2 x L)/(24 x n^2 x P^2)

- w = weight of tape per metre, W = total weight, n = number of equal spans. Sag correction is **always negative** .
- Sag is reduced by increasing the pull or the number of supports. Normal tension: correct pull P = 0.204 x W x sqrt(AE) / sqrt(Pm - Po) (for no-sag case).

(e) Correction for slope

Cslope = h^2 / 2L (approx.) or L (1 - cos theta)

Horizontal distance D = sqrt(L^2 - h^2) = L cos theta

- h = difference in level between the two ends, L = slope distance. Always **negative** .

(f) Correction for alignment, (g) correction to MSL

Calign = d^2 / 2L (d = deviation from line)

Cmsl = - L x h / R (h = mean elevation, R = 6370 km)

MEMORISE:

Sag, slope and alignment corrections are always SUBTRACTIVE. Temperature and pull may be + or -, wrong length depends on whether the tape is long or short.

Worked example 1

A 30 m chain was found to be 0.05 m too long. A line measured 450 m. Find true length.

True length = 450 x (30 + 0.05)/30 = 450.75 m

Worked example 2 (slope)

A line of 120 m on a uniform slope has a height difference of 4 m. Find horizontal distance.

C = h^2/2L = 16/240 = 0.067 m

D = 120 - 0.067 = 119.933 m

Errors in chaining (summary)

- **Personal errors** - wrong reading, careless marking, bad ranging.
- **Natural errors** - temperature, wind, sag, slope.
- **Instrumental errors** - wrong length, bent links, loose rings.
- Mistakes: miscounting chain lengths, misreading tallies, wrong booking of figures.
- Remedies: standardise the chain before use, check it with a test gauge, apply corrections.

14. Prismatic Compass

A ***prismatic compass*** is a portable magnetic instrument used to measure the bearing of a line. It has a magnetic needle on an agate cap, graduated aluminium ring (card) and a prism to read the bearing.

Parts

- ***Box*** (non-magnetic), with glass cover and lid; ***magnetic needle*** with agate cap on a pivot.
- ***Graduated ring*** (0 to 360 deg, graduated inverted; 0 at the south end).
- ***Object vane*** with hair, ***eye vane*** with slit and ***prism*** , sunshades (coloured glasses), lifting pin and brake pin.

Temporary adjustments

- ***Centring:*** place the compass over the station using plumb bob.
- ***Levelling:*** keep the box horizontal so that the card swings freely (ball and socket).
- ***Focusing the prism:*** move prism up or down until the graduations are sharp.

Observation

Sight the object vane to the ranging rod. Look through the slit, read the graduation seen against the hair line (through prism) when the needle stops.

15. Bearings and Included Angles

- ***Meridian:*** fixed reference direction – true, magnetic or arbitrary.
- ***True meridian:*** line through geographical North Pole and South Pole. True bearing = azimuth.
- ***Magnetic meridian:*** direction of freely suspended magnetic needle. Magnetic bearing.
- ***Arbitrary meridian:*** any convenient direction (e.g. the first line of the survey).
- ***Grid meridian:*** direction parallel to the central meridian of a grid system.

Bearing

Bearing of a line is the horizontal angle between the meridian and the line, measured clockwise from the North.

- ***Fore bearing (F.B.):*** bearing taken in the direction of progress of the survey.
- ***Back bearing (B.B.):*** bearing in the opposite direction.

B.B. = F.B. +/- 180 deg
(add 180 if F.B. < 180; subtract 180 if F.B. > 180)

Included angle

Interior angle between two successive lines at a station.

Included angle = F.B. of next line - B.B. of previous line
(add 360 deg if the result is negative)

CHECK:
For a closed traverse of n sides: sum of interior angles = (2n - 4) x 90 deg. Check this to find observation errors.

16. Systems of Bearing: W.C.B. and Q.B.

Whole Circle Bearing (W.C.B.)

Bearing measured **clockwise from North** from 0 to 360 deg. A single value fixes the line. Used in the prismatic compass.

Quadrantal Bearing (Q.B.)

Bearing measured from North or South, **whichever is nearer** , towards East or West, with value from 0 to 90 deg. Used in the surveyor's compass. Written as N 40 E or S 25 W.

W.C.B. range	Quadrant	Q.B. formula
0 - 90	I (NE)	N (W.C.B.) E
90 - 180	II (SE)	S (180 - W.C.B.) E
180 - 270	III (SW)	S (W.C.B. - 180) W
270 - 360	IV (NW)	N (360 - W.C.B.) W

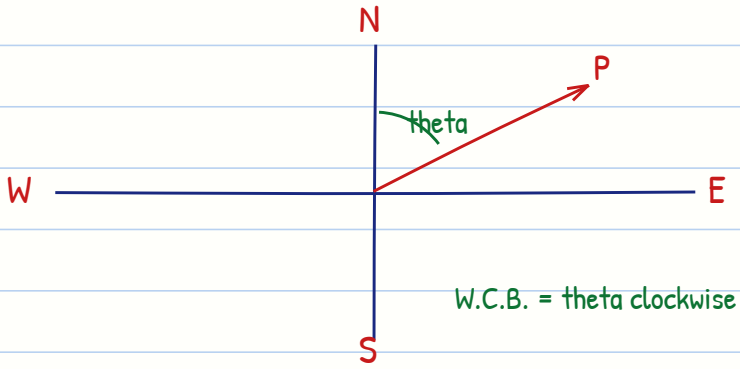


Fig 4: whole circle bearing

Examples

- F.B. of AB = 125 deg 30' -> B.B. = 305 deg 30'. Q.B. of AB = S 54 deg 30' E.
- W.C.B. 245 deg -> Q.B. = S 65 W. W.C.B. 320 deg -> N 40 W.
- Q.B. N 30 W -> W.C.B. = 330 deg; S 20 E -> W.C.B. = 160 deg.

17. Local Attraction

The magnetic needle gets deflected from the magnetic meridian due to nearby magnetic substances - iron objects, steel structures, rails, cables, magnetite rocks. This is ***local attraction*** .

- Detection: if F.B. and B.B. of a line differ by exactly 180 deg, both stations are free from attraction. Otherwise, there is local attraction at one or both ends.
- Correction methods: (1) start from a line whose F.B. and B.B. differ by 180 deg and take that as correct; (2) distribute the error among the lines using the included angle method.
- ***Method 1:*** the observed bearings of lines at an affected station are all wrong by the same error; correct by applying the same error.
- ***Method 2 (angle method):*** the included angles at all stations are not affected by local attraction. Compute them and correct the bearings of all lines from a correct bearing.

Sample

FB of AB = 60 deg 30', BB = 240 deg 30' (difference 180). So A and B are free. FB of BC = 120 deg 10', BB of CB = 301 deg 40'. The difference is 181 deg 30', so C has attraction: correction at C is -1 deg 30' applied to all bearings measured at C.

18. Magnetic Declination and Dip

Magnetic declination (D)

Horizontal angle between the true meridian and the magnetic meridian. It varies with place and time (secular, annual, diurnal and irregular variations).

- ***East declination*** – magnetic north lies to the east of true north (+ve). ***West declination*** – magnetic north to the west (-ve).

> True bearing = Magnetic bearing + declination
> (declination East = +, West = -)

- Lines of equal declination = ***isogonic lines*** ; lines of zero declination = ***agonic lines*** .
- Variations: secular (long-term, ~ 100–200 yr), annual (about 1–2 min), diurnal (changes through the day, max about 1 deg), irregular (magnetic storms).

Dip of the needle

The angle that the magnetic needle makes with the horizontal at a place. The needle's north end dips down in the northern hemisphere and the south end in the southern hemisphere.

- Dip is 90 deg at the magnetic poles and 0 deg at the magnetic equator. To balance it, a ***rider (small brass coil)*** is placed on the needle's higher end.
- Position of the rider shows which hemisphere the survey is in. This is done by the manufacturer for each country.

19. Quick Revision Points

- Whole to part, two independent measurements.
- Corrections: sag, slope, alignment always subtract.
- B.B. = F.B. +/- 180. Included angle = FB(next) - BB(previous).
- W.C.B. 0–360 from North; Q.B. 0–90 from N/S towards E/W.
- Local attraction: FB – BB difference not equal to 180 deg. Declination: true = magnetic + declination (E +).